

How to set-up and maintain a successful Open Data Initiative

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Open Data

What? Why? Who? How? When?

- What is **Open Data** and an **Open Government Data** (OGD) initiative.
- **Steps and process** to implement a successful OGD initiative.
- Potential **impact** of an OGD initiative.
- How to **engage** with the different stakeholders.

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What is Open Data?

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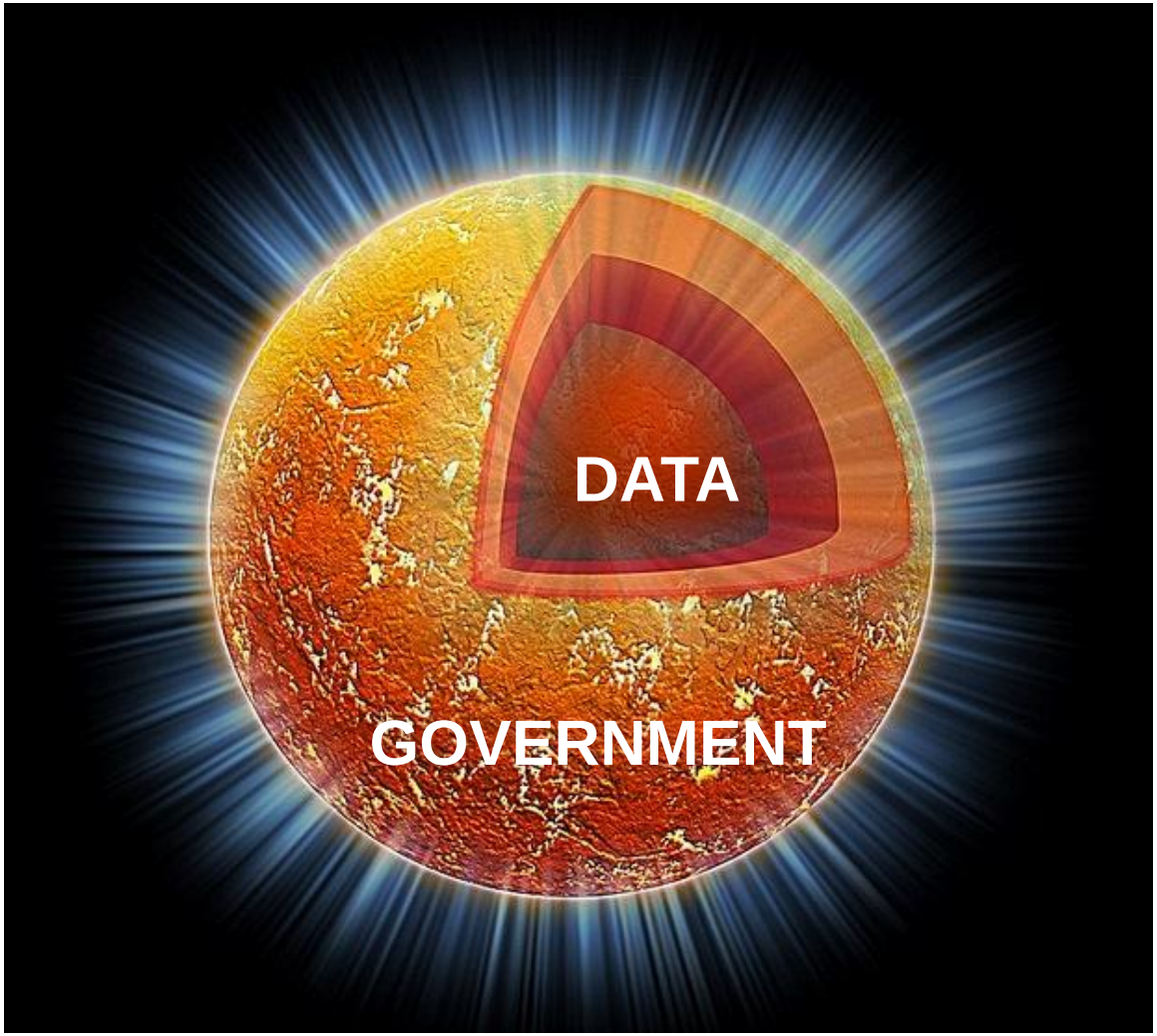
Definition



"Expose (raw) data in **free open standard formats** and under **non-restrictive licenses** to make it accessible to all and allow **reuse**"

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Definition



http://www.flickr.com/photos/28634332@N05/5473561903/

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8+2 Principles



Completeness
Primacy
Timeliness
Easy Access
Machine readability
Non-discrimination
Non-proprietary
License-free
Permanence
Usage cost

http://www.opengovdata.org/home/8principles
http://sunlightfoundation.com/policy/documents/ten-open-data-principles/

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Benefits



“Data is the new oil for the digital age” - Neelie Kroes, Vice-President of the European Commission responsible for the Digital Agenda.

http://europa.eu/rapid/press-release_SPEECH-12-149_en.htm

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Benefits



Information is power

- New technologies are making it easier to **collect** data, **share** information, target resources, provide **feedback** and **measure** progress.
- Information can help to **build trust**, allowing people to exercise their rights and make more informed choices in their daily lives.

http://www.devinit.org/wp-content/uploads/Turning-information-into-action-A-post-2015-agenda-for-ending-poverty-Jan-2013.pdf

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Benefits



☑ **Transparency** and Accountability.

☑ Public **participation** and Inclusion.

☑ Business opportunities, **Economic growth** and Employment.

☑ Data quality and **Interoperability.**

☑ **Efficiency** and better Public Services

☑ **Social value** and Benefits.

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Benefits



Bottom line

Innovation (Jobs and growth)

Efficiency (Community)

Transparency (Governance)

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Barriers



Economic: it will cost too much, there are no resources available or may lose revenue generated by certain data.

Capacity: challenges when reviewing, releasing and maintaining open data.

Organisational: requires coordination across multiple agencies and the diverse stakeholders community.

Cultural: Excessive government secrecy, or general resistance to change.

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Barriers



Legal: Legal or privacy issues.

Expertise: Absence of digital literacy or specific knowledge on the matters.

Data availability: Data is generally bad quality or non existent.

Technological: Lack of proper infrastructures.

Lack of interest: Not being considered necessary.

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Barriers, an example:



Government's (excessive) secrecy

"If people don't know what you're doing, they don't know what you're doing wrong" – Yes Minister on Open Government (1980)

http://www.yes-minister.com/index.html

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Barriers



Bottom line

It is **tough**

It is **expensive** and there is no ROI

I'm **not required** to do it

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The data hugging syndrome



Symptoms

Loss of control: Authenticity, provenance, falsification, quality problems

Sense of danger: Legal challenges, Privacy and National security, unwelcomed exposure

Overwhelming complexity: Capacity building and procedural changes required

Bankrupt panic: Investment, ROI, Loss of licence revenue, Customer service

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Myths and Misunderstandings



- We **do not own the data**, it is held separately by several different organizations.
- It is **technically impossible**, the data is too large to be published and used.
- We know the **data is wrong** and people will tell us where it is wrong. That will make people angry.
- People will draw **superficial conclusions** and construct *league tables* from data ignoring the wider picture.
- It might be combined with other data to identify individuals/**sensitive information**.
- It will **cost too much** to put it into standard format. Our IT suppliers will charge us a fortune to so.

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Myths and Misunderstandings



All these have some truth in them

- They rationalise the official ***fear of the unknown***.
- Data managers need to be helped through.
- Need to build a **balanced ecosystem.**
- **Examples and precedents** are your friends.
- For first phase, **compromise** then scale.
- Release **data that matters.**

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Positives outweigh concerns

Evidence: Economic growth

- OGD in EU would **increase business activity of €40bn**. Indirect benefits up to **€140bn/year**.
- **+5000 jobs and €500m benefits** in Spain from PSI reuse.
- **€62m benefits** (2005-2009) and **€2m cost** (2002-2009) in Denmark. **€14m benefit against €0.2m cost** in 2010.

http://ec.europa.eu/information_society/policy/psi/docs/pdfs/report/psi_final_version_formatted.docx
http://www.ontsi.red.es/ontsi/sites/default/files/121001_red_007_final_report_2012_edition__vf_en_1.pdf
http://epsiplatform.eu/content/value-danish-address-data

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Positives outweigh concerns

Evidence: Data quality

- UK released data on location of nearly 300.000 bus-stops.
- Community **corrected 18.000 of them** (6% approx..) thanks to *Open Street Maps*, improving official data accuracy.

http://www.gong.hr/download.aspx?f=dokumenti/20120928-zagreb-conference-1.0-AndrewStott.ppt

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Positives outweigh concerns



Evidence: Interoperability

- Open Data British Columbia, CA.
- Built a provincial open data portal.
- As about **one third of all the traffic originates from government.**

http://openspending.org/resources/gift/chapter2-2.html

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Positives outweigh concerns

Evidence: Social value

- UK published **comparable data** on the individual clinical outcomes in 2004.
- Seven years later there are **1.000 fewer deaths** in heart surgery units each year.

http://10years.reform.co.uk/essays/The-future-is-Open-why-transparency-will-be-the-organising-principle-of-21st-century-public-services.pdf

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Positives outweigh concerns



Evidence: Transparency

- Capitation funds reaching schools in Uganda **increased from 20% to 90%** between 1995 and 2001.
- Thanks not only to **public expenditure tracking surveys** and **data publication**, but also to concurrent reforms in Uganda's education and fiscal systems.

http://www.cgdev.org/files/15050_file_Uganda.pdf

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Positives outweigh concerns

Evidence: Better public services

- Apps for democracy **$50k investment.**
- 47 Phone, Social Networks and Web apps in 30 days (a **$2.3m value - ROI 4000%**)
- 1-2 years time estimation to complete the procurement process and receive apps.

http://www.appsfordemocracy.org/citizen-engagement-through-apps-for-democracy-community-edition/

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Positives outweigh concerns



New business opportunities

Data analysis

Data mining

Data combination

Visualisation

Niche markets

…

If it is hard for governments to experiment then **let others experiment for you.**

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State of the art 2009: 22



http://datos.fundacionctic.org/sandbox/catalog/faceted/

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State of the art 2010: 58



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State of the art 2011: 140



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State of the art 2012: 202



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State of the art 2013: 217



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Open Government Partnership: 58



http://www.opengovpartnership.org/countries

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LMICs: 217



https://maps.google.de/maps/ms?msid=214949500913461139756.0004d6af336a205394ed1&msa=0

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Africa: 16



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Millennium Development Goals



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MDGs

An honest and responsive government

This means that governments should be effective, transparent, accountable and not corrupt. People should have a say on what the government's priorities should be, and confidence that they will implement those priorities competently. Governments should agree and implement standards for making information available to all people on how public money is spent

MY WORLD.
THE UNITED NATIONS
GLOBAL SURVEY
FOR A BETTER WORLD.

Almost as difficult as choosing the right goals, is to find adequate measures that will mark progress toward them.

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MDGs

Open, accountable and inclusive Governments

- Openness in the design of the post-2015 framework.
- Openness in monitoring of investment and outcomes.
- Openness in terms of making that information widely available and accessible.
- Making relevant data accessible to citizens and accountability institutions.
- Opportunities for their citizens to engage.

https://s3.amazonaws.com/one.org/images/ONE_HLP_Report_-_FINAL.pdf

http://www.savethechildren.org.uk/sites/default/files/images/Ending_Poverty_in_Our_Generation_Africa.pdf

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The components of an Open Data initiative

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Objective



Development and deployment of local sustainable Open Data initiatives

Country-based approach: implementation of complete OGD initiatives to build local capacities

Focus not only on data release but also on **creating ecosystems** for the use and re-use of the released data by other actors such as civil society, academia, business, media or any other organisations that may benefit of them.

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Open Government pioneers

How it all began?



Most of the things were already in place (e.g. political willingness, IT infrastructure, capacity etc.)

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Hypothesis



What if we use the knowledge in Western world Open Data projects in LMICs?

- It wouldn't work out of the box
- Some things will be applicable "as is"
- Some will need little or huge changes
- Some new challenges arise (and will need to be addressed)



http://www.webfoundation.org/projects/open-government-data/

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Need to go one step back

- What is needed?
- How much of what we know from existing initiatives is applicable?
- What are the indicators that will enable the definition of OGD readiness in a given country?

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LMICs' singularities



- **Transparency and accountability** to encourage foreign aid and investments
- **ICT potential to provide basic services** (health, education, business, government...) to rural communities and under-privileged populations
- **Affordability** (e.g., tools, assistive technologies)
- **Pronounced effects of some barriers** (e.g. age, literacy, language, experience)
- **Integrate with current communications** (business and social groups, radio, TV, SMS ..)

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Dimensions



Political

Technical

Legal

Economic

Organizational

Social

Data

Stakeholders

Methodology: http://www.webfoundation.org/projects/open-government-data-feasibility-studies/
https://data.worldbank.org/sites/default/files/1/od_readiness_-_revised_v2.pdf

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Socio-economic



- Demographics
- Labour market
- Quality of life
- Education and literacy
- Macroeconomic stability
- Financial markets
- Markets for goods and services
- Business environment

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Institutional framework



- Leadership
  - Is there a **political top-level leadership** to facilitate an Open Data initiative?
- Sustainability
  - Is there a **management structure** that facilitates Open Data?
  - Is the Government **middle layer ready** to facilitate an Open Government initiative?

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Institutional framework



- Policies and legal framework
  - Does the country have **legislation** related to PSI reuse, data openness, transparency, official secrecy and/or privacy? (e.g. FOIA, Privacy Act)
  - Does the country have a **licensing and copyright framework**?
  - What policies/laws help or hinder the **use of information by civil society**?

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Institutional framework



- Economics
  - Are there sufficient **resources in place to fund an initial phase of an Open Government Data** initiative and support the necessary infrastructures and skills needed?
  - To what degree and how is the expense of opening significant amounts of government data likely to be an issue?

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Technology and Infrastructures



- What is the **importance of ICT** to Government?
- How developed are **electronic services** in the country? Is there any **Interoperability Framework**?
- What is **ICT Development level and evolution**? (e.g. connectivity, mobile coverage, etc.)

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Government Data



- What level of **data collection** does the government undertake?
- Are there **inventories of data** held at government?
- What is the **quality** of data?
- How easy is to **reuse** available data?
- What are the **most valuable government datasets** when publicly available?

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The Open Data Ecosystem



“It has to happen at the top, it has to happen in the middle and it has to happen at the bottom” - Tim Berners-Lee, inventor of the Web.

http://www.opensocietyfoundations.org/sites/default/files/open-data-study-20110519.pdf

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The Open Data Ecosystem

Several Actors

- **Political Level:** to achieve top level political willingness.
- **Public Administration Level:** to raise awareness, improve skills, make it sustainable.
- **Civil Society Level:** to realize the value behind the data and keep government opening up.

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Joy's Law

"No matter who you are, most of the smarter people work for someone else" – Bill Joy, Sun Microsystems co-founder.

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Collorary: The many minds principle

“The coolest thing to do with your data will be thought of by someone else” – Jo Walsh and Rufus Pollock, OKFN.

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Open Data stakeholders and roles

- **Public sector** (political guidance, top level direction, guide implementation, policy-making, agenda-setting, implementation and support)
- **Civil Society Organisations** (use Open Data for their advocacy work, monitoring policies and encourage political participation)
- **Civic hackers** (apply technological skills to make sense of data)

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The Open Data Ecosystem



Open Data stakeholders and roles

- **Media** (use Open Data in reporting and data journalism)
- **Academia & Research** (use Open Data in research, provide theoretical concepts and frameworks)
- **Private sector** (build business on Open Data and support infrastructures)
- **Donors, Foundation & Int. Orgs.** (funding, knowledge transfer and capacity building)

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Community engagement

- Are there Public Sector Information **re-use outreach and encouragement** activities?
- Does a **network or community** exist that bridges the gap between the middle governmental layer and civil society?
- Are there donors active in the country that could be useful **allies**?
- How dynamic is the media sector?

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Community engagement

- Is there an **academic or research community** that both trains people and has people skilled at data science?
- Are there **data reuse initiatives or champions** from the civil society?
- Is there a **potential user base** that may make use of targeted data?
- How dynamic is the ICT sector?
- Is there a **data economy** already in place?

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Lessons learned?



“Open Data must have a well balanced ecosystem” – Nigel Shadbolt, Head of the Web and Internet Science Group at the University of Southampton.

http://www.deloitte.com/assets/Dcom-UnitedKingdom/Local%20Assets/Documents/Market%20insights/Deloitte%20Analytics/uk-insights-deloitte-analytics-open-data-june-2012.pdf

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Lessons learned?



Quick OGD portal vs. sustainable long-term OGD initiative

- OGD Portal should be just a consequence **not an end in itself**.
- Technical approach vs. **OGD ecosystem** (actors and dimensions).
- Start simple but with **long term goal in mind**.

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Lessons learned?



- The issue with **Open**
  - Several times not really Open (have a look at the licenses)
- The issue with **machine-readable, standard formats**
  - PDFs, Excel, etc. (documents not data)
  - Web Architecture ≠ Web as file server.

http://opendefinition.org/licenses/

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Lessons learned?



- **Raw data** now… and **better data** afterwards (make it work and then make it better)
  - Start with the *low hanging fruit* and improve over time (make it work an then standardize)
- Do not try to enforce an **specific architecture**
  - Chances are you could not deploy it.
  - Try to adapt to existing systems and build on top of them as a start.

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How to set-up an Open Government Data initiative?

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Steps



5 fundamental steps

1 – **Organise**

2 - **Standardize**

3 – **Publish**

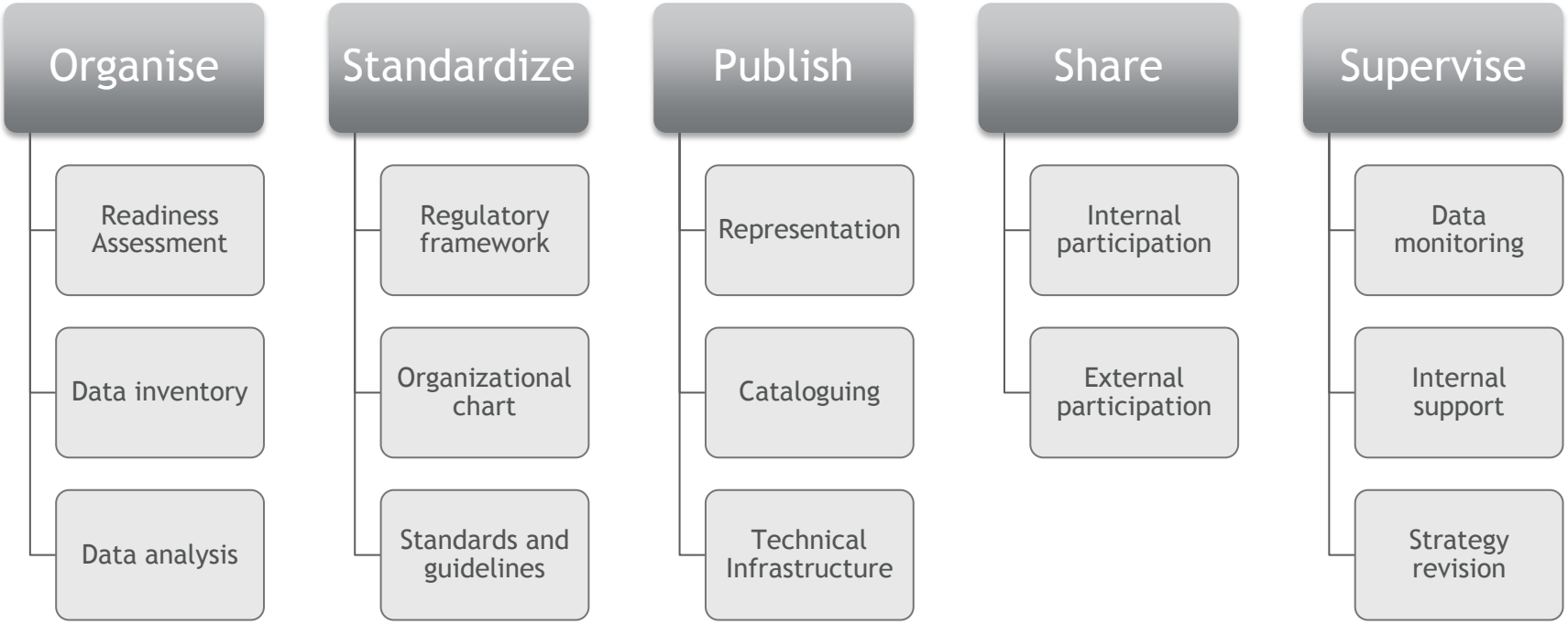
4 – **Share**

5 - **Supervise**

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Steps





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Organise



What data am I currently managing and/or publishing?

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Data inventory

- **Health and welfare** (e.g. public health inspections, hospitals performance, etc.)
- **Education and training** (e.g. school performance, educational resources, etc.)
- **Public finances and procurement** (e.g. national budget, taxes distribution, contracts, etc.)
- **Natural resources and environment** (e.g. extractive industry, pollution, etc.)
- **Geospatial** (e.g. maps, points of interest, etc.)

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Data inventory

- **Politic and Policies** (e.g. laws, official proceedings, bulletins, election results, etc.)
- **Justice and Public safety** (e.g. crime data)
- **Public directories** (e.g. addresses and contact for schools, hospitals, libraries, police stations, etc.)
- **Transportation** (e.g. information about roads and public transportation)
- **Statistics** (e.g. socio-economic and demographic information)

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Organise



“If I had asked people what they wanted, they would have said faster horses” – Henry Ford, Founder of the Ford Motor Company.

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Data analysis

Identification of datasets (*data cards*)

What: datasets, descriptions, formats, volume, etc.

Where: location, storage, distributions, etc.

Who: data owners and managers, roles, contacts, etc.

When: data age, updates frequency, creation date, expiration date, etc.

How: means of access, grants and restrictions, terms of use, quality, etc.

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Standardize



How could I structure data in a way that it can be easily reused?

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Regulatory framework

- **Open by default** (keeping logical privacy and security restrictions)
  - Free and unlimited access, no justification required.
  - Ensure permanent availability in the future.
- **Terms of use**
  - **Licensing**: free or open (at least use, reuse and distribute), simple and unified, self-described and machine readable, etc.

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Regulatory framework

- **Legislation**
  - **Laws** and **policies** (FOI, PSI re-use, standards, quality, etc.)
  - **Security** and **privacy** (be careful with the new *mash-up* possibilities)
  - Keep also into consideration **other applicable frameworks** (regional, national, international, specific, etc.)
- **Services procurement**
  - Same requirements for suppliers and subcontractors (*openness clause*)

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Organizational chart

- **Management**
  - General management.
  - Internal administration (data publishers)
  - External administration (data consumers)
- **Coordination**
  - Promoters and different public bodies.
- **Communication**
  - Dissemination and outreach.
- **Technical team**
  - Content management, developers, system admin.

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Standards and guidelines

- **Metadata definition** for datasets description and characterization to facilitate findability and reuse.

Property	Description	Data type	Multiple	Required
Name	Name given to the directory.	Short text string.	N	Y
Description	Brief narrative summary of the directory.	Text string.	N	Y
Homepage	The homepage associated to the directory	URI for the directory homepage.	N	Y
Language	The language or languages used in the textual metadata for describing items in the directory.	Language code as defined by BCP47[10] and recorded at the IANA registry[11].	Y	Y
Themes	The full set of subjects and topics that are addressed by the directory.	Predefined texts from the set of Subjects and Topics specified by the directory taxonomy[12].	Y	Y
Geographic coverage	The geographical area covered by the directory.	Predefined texts from the set of Territories specified by the directory taxonomy.	N	N
Publisher	The entity or entities responsible for making the directory online.	Text string or a reference to them.	Y	Y

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Standardize



Standards and guidelines

- **Vocabularies** to describe and classify all the records of the catalogue in common terms.
- **Taxonomies definition** to provide standard names and references for common objects.
- **IDs scheme** to maintain persistent identifiers and preserve data over time.
- **Catalogue** requirements and functional specification.

Always Document!

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Catalogue functionality

- **Informative sections** (about, help, terms of use, news, blogs, stats…)
- **Data Sets Catalogue** (data catalogue, semantic capabilities…)
- **Data Sets Services** (browsing and searching, querying, download, apps, visualizing…)
- **Community Services** (communities of interest, requests, comments…)
- **Back End** (user management, data management, applications management…)

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Publish



How should I share the data?

http://www.w3.org/TR/gov-data/

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Publish



“There wouldn’t be Open Data without the Web”

- Need to **be inclusive**, minimize/avoid a data divide… address the challenges!
- Need to provide global leadership across dimensions to **advance the Web and Open Data**.
- Need to act as catalyser and hub to **make the Web available to all**.
- Think of the Web (technologies) as a mean, **deliver as needed** (community radios, mobile phones, etc.)

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Publish



“Keep it simple. When in doubt during design, choose the simplest solution”– Architecture Principles of the Internet.

http://www.ietf.org/rfc/rfc1958.txt

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Metadata Standards

- **Re-use** of **standardized mainstream vocabularies**: DCAT, ADMS, Dublin Core, vCard, FOAF, iCal…
- In order to **maximise interoperability** and **avoid ambiguities**, use also **uniform formats** to specify metadata (i.e., ISO-8601 to represent dates)

http://www.w3.org/TR/vocab-dcat/
http://www.w3.org/ns/adms
http://dublincore.org/
http://en.wikipedia.org/wiki/VCard
http://www.foaf-project.org/
http://en.wikipedia.org/wiki/ICalendar

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Taxonomies and Vocabularies

- Existing **reference vocabularies** should be first evaluated in order to establish those that are the most appropriate: Data Cube, the eGov Core Vocabularies…

  http://www.w3.org/TR/vocab-data-cube/
  https://joinup.ec.europa.eu/community/core_vocabularies/description

- The SKOS standard is a data model for **sharing and linking** through the Web's basic structure and content.

  http://www.w3.org/TR/2009/REC-skos-reference-20090818/

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Publish - Representation

IDs Scheme

*http://{base}/**{type}[/{domain}][/{sector}][/{resource}]***

- The ability to identify resources via URIs is **one of the foundations of the Internet**.
- If we provide **permanent** URIs that are **self-describing** and follow established patterns; it will be much easier to find the desired information.
- The usage of these **persistent identifiers** will also provide **reliable references to the data**, a basic requirement to ensure information access.

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Publish - Representation



- Use **URIs as names** for things.
- Use HTTP URIs so that people can **look up the names**.
- When somebody looks up an URI it should **provide useful information** using standards.
- Provide at least one **machine-readable** representation of the resource identified by the URI.
- Use **consistent**, **extensible** & **persistent** URIs.
- Create URIs that are **understandable** and **meaningful**.
- Never expose information about the technical implementation of URIs.
- Include links to other URIs so that people can discover more **related things**.

http://www.w3.org/DesignIssues/LinkedData

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Updates and versioning

- How to **preserve integrity** of published data is one of the **key challenges** (new resources, new links between data sets, obsolescence and deletion of old material…)
- Data **persistence must be ensured** over time if we expect the stakeholders' community to re-use it.
- Keep track of **relationships between different data sets versions.**
- **Document changes** that have been carried out, including appropriate **version metadata**.

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Publish - Cataloguing

Data Catalogues

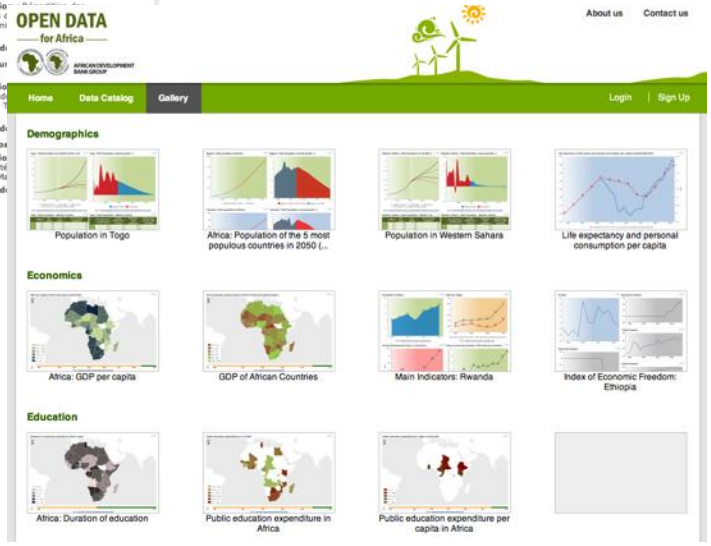












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Publish - Cataloguing



The 5 ★ of Linked Open Data

★ data on the web

★★ structured data

★★★ non-proprietary formats

★★★★ URLs to identify things

★★★★★ link to other's data

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Publish - Cataloguing

The road ahead

Linked Data

RDFa

APIs

XML

RSS/atom

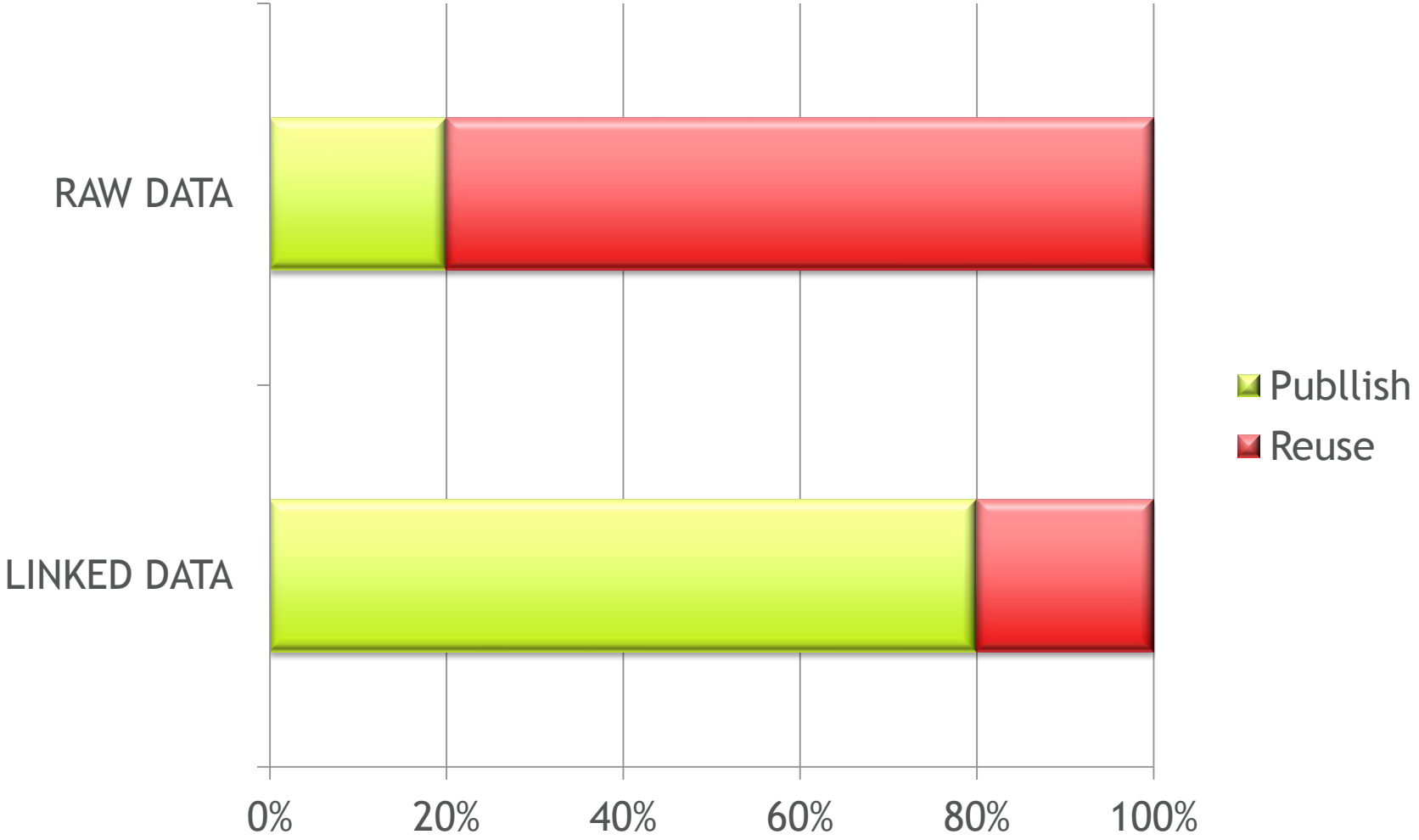
HTML Scrapping

Raw Data



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Publish - Cataloguing



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Publish - Infrastructure

◇	A
1	178226,05,058496,,2,10/20/2008,12/31/2008,05/21/2009,Y,N,13,00450,4,05/18/2009,N,05/20/2009,,04/30/2009
2	178500,5,679003,,2,10/01/2008,12/31/2008,05/27/2009,N,N,13,00380,4,05/20/2009,F,05/22/2009,,05/15/2009
3	178784,05,148021,,2,11/07/2008,12/31/2008,05/29/2009,Y,N,13,00380,4,05/28/2009,L,05/28/2009,,05/19/2009
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5	179142,05,109115,,2,11/03/2008,12/31/2008,06/04/2009,Y,N,13,00380,4,05/28/2009,N,06/09/2009,,05/21/2009
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7	180007,05,743183,,2,10/06/2008,12/31/2008,06/11/2009,Y,N,13,00380,4,06/05/2009,N,06/10/2009,,05/26/2009
8	180168,3,679619,,2,10/31/2008,12/31/2008,06/13/2009,N,N,13,00380,4,06/08/2009,F,06/12/2009,,05/29/2009
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14	181408,05,447574,,2,10/28/2008,12/31/2008,06/18/2009,Y,N,13,00380,4,06/09/2009,L,06/16/2009,,05/29/2009
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Publish - Infrastructure

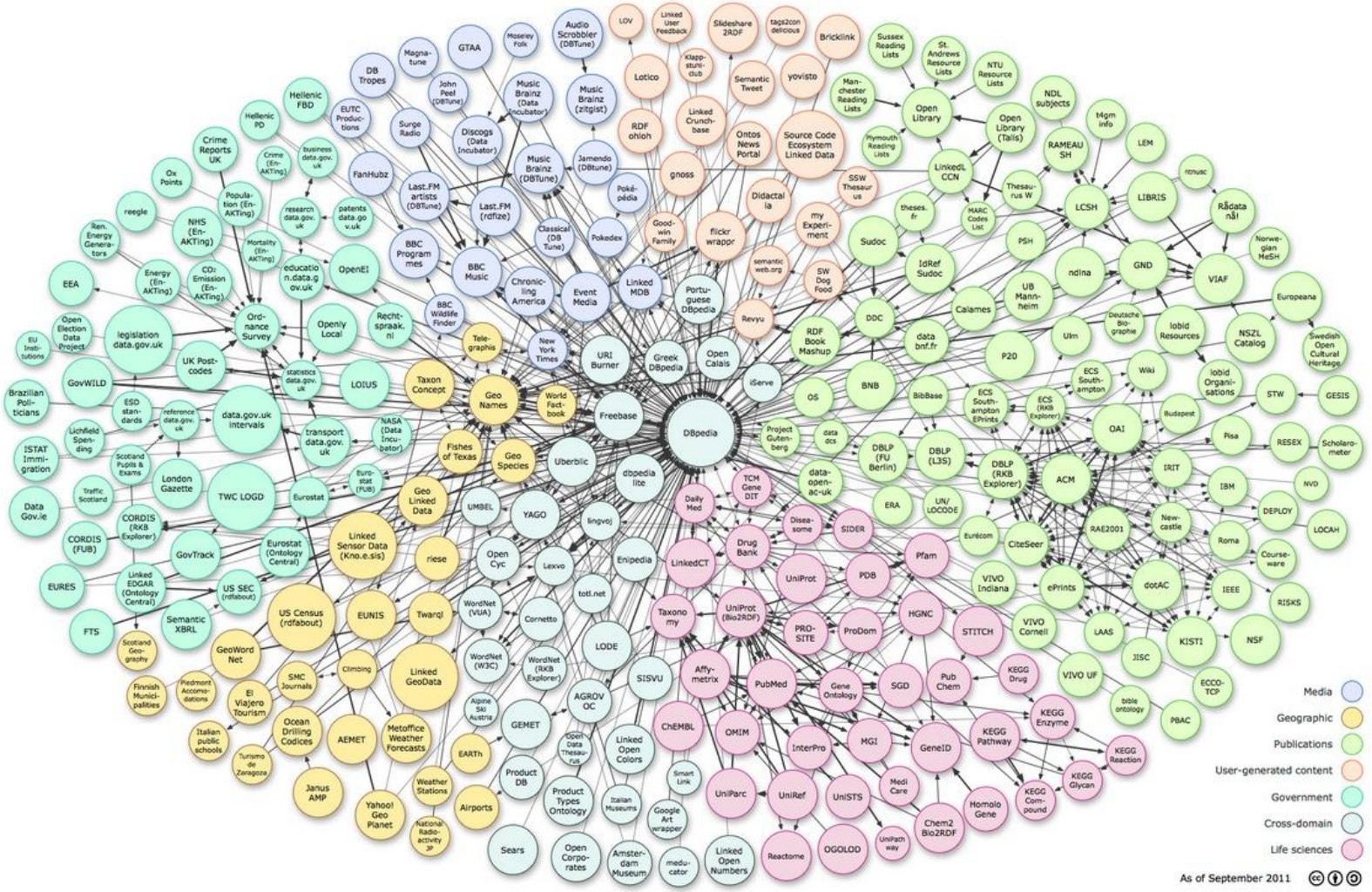
Raw data vs. machine readable





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http://lod-cloud.net/

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LOD benefits are great, but…

- Modelling is **though**.
- Good to combine multiple sources, but still there are **other issues** (licensing, data quality, provenance, trust…)
- Sometimes *slow* performance.
- **Better toolkits** are needed.
- **Capacity building**.

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What platform?

- **Data Catalog Interoperability** (allow catalogues federation with standard schema – DCAT)
- **Data Portability Based** (use standard data formats such as JSON, XML or CSV, as well as RDF and other Linked Data standards)
- **Application Portability** (based on Open Data API Standards, such as REST, HTTP, and Structured Query Language (SQL)





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“Tim Berners-Lee didn’t develop hundreds of millions of websites. He designed a platform so that others could” – Tim O’Reilly, Founder of O’Reilly Media.

http://www.businessofgovernment.org/report/designing-open-projects-lessons-internet-pioneers

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Share



How could I be more open?

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Internal participation (Government)

- The consolidation of an Open Data initiative is in need of involvement from all Government bodies to achieve **full cultural integration** with internal processes.
- **Resistance to change** is a delicate issue.
- Governments tend to **avoid risks**.

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Internal participation (Government)

- **Attract new public bodies**
  - Project presentation and communication.
  - Encourage participation.
- **Raise awareness**
  - Internal meetings, workshops, seminars…
- **Capacity building**
  - Open Data, benefits, examples…
  - Data processing and classification.
  - Data catalogue inner workings.

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Share (externally)



External participation (Other Stakeholders)

Open Government Data initiatives should be seen as **collaborative projects** within a **changing environment**, where **the end may not be clear** and **the *coordinating entities* do not have all the resources to create change needed** by themself.

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Share (externally)



“If you prevent people from doing bad things, you prevent also them from doing good things” – Jimmy Wales, Wikipedia co-founder.

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Share (externally)



10 lessons from Internet pioneers

- 1 - Let **everyone** play: bring in additional resources, attracts advocates, and captures new ideas.
- 2 - Play **nice**: a common mistake is reduce participation by setting high walls to protect from low-quality contributions, vandalism, etc.
- 3 - Tell what you are doing **while you are doing it**: improve understanding and contributions, avoid overlap and create adjacent value, build trust.

http://www.businessofgovernment.org/report/designing-open-projects-lessons-internet-pioneers

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Share (externally)



- 4 - Use **multiple communication channels:** do not focus only on the Web.
- 5 - **Give it away:** governments shouldn't be proprietary about what they know and learn.
- 6 - Reach for the **edges:** need observations and insights from the edges when we face complicated problems.
- 7 - Take advantage of **all organizations:** OGD is likely to require contributions from an ecosystem of different participants with varying interests and incentives.

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Share (externally)



- 8 - Design for **participation:** breaking work into manageable pieces makes it easier to understand what needs to be done and become involved.
- 9 - Increase **network impact:** a community model based in a network of nodes that can interact back and forth will create an exponential growth.
- 10 - Build **platforms:** help people and organizations coordinate their activities so they are jointly more productive.

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Dissemination and outreach



- **Public consultation:** ask people what data they want.
- **Round tables and networking:** share experiences face to face and set common goals.
- **Communities of interest:** gather together people with different motivations and profiles, but that share interest in a given topic.
- **Communities of practice:** e.g. dissemination and engagement, data standards and homogenisation, data catalogues best practices, etc.

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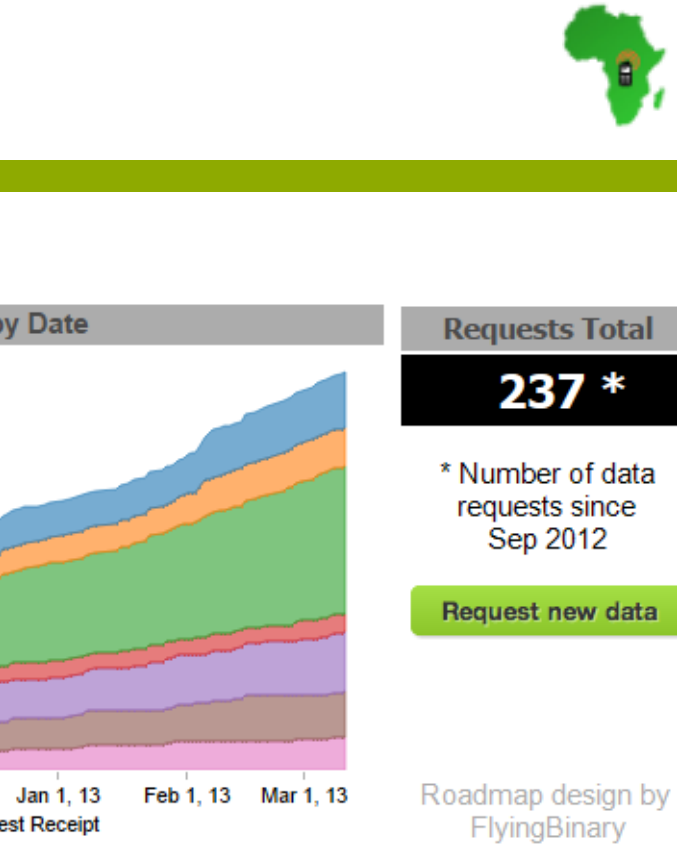
Dissemination and outreach



- **Public-private collaboration:** explore possible synergies between government and other external entities, but avoiding exclusive alliances.
- **Dissemination** (online and offline): news, blogs, social networks, press releases, multimedia, etc.
- **Education and investigation:** training of enough human resources with skills to take advantage of OGD.
- **Entrepreneurship promotion:** capacity building, support, business models, challenges, Hackatons, etc.

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Supervision



http://www.data.gov.uk/odug-roadmap

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Data monitoring

- **Data catalogue** monitoring: visitors, sources, keywords, bounces, time on site, devices, locations, etc.
- **Data sets** usage and characteristics: new datasets, updated datasets, publishers, formats, licenses, topics, apps, data requests, etc.

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Internal support

- Knowledge **transfer**.
- **Assistance** for public bodies.
- **Procedural** support for OGD processes.
- **Technical** support for OGD tools.
- **Capacity** building.
- Guidelines and educational **resources**.

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Strategy revision

- Have a look back to the **original objectives**.
- Detect **achievements**, **failures**, **mistakes**, **best practices**, **champions**, etc.
- **Analyse** and **evaluate** the results.
- Share the **indicators**.
- **Update** the strategy accordingly.

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Some final remarks

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Final remarks



Spend some time to **analyse the starting point**, identify champions and collaborate, but…

Get started now! you can start small and simple, then grow steadily.

Address fears and misunderstandings from the very beginning.

Share and engage data has no value if it is not being used, use examples that focus on everybody's live improvement and avoid the "cool visualization" effect.

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Final remarks



Technology is a small percentage: but still remember there are 3 fundamental steps

"**Identify** the data that you manage, **represent** that data in a way that people can use it and **expose** the data to the wider world" – Jeni Tennison, UKODI.

http://www.jenitennison.com/blog/node/100

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That’s all

Thank you for your attention!

Any other Questions?

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Do you need some support while Opening up your Data?

Just keep in touch

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es.linkedin.com/in/carlosiglesiasmoro/en

This presentation: http://tinyurl.com/eGovAfricaOGD

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